

VIKALP — Settlement Evidence Assessment Report

This report is not a relocation order, disaster declaration, government approval, evacuation order, autonomous AI decision, or final risk certification.

Report ID	VIKALP-RPT-0001-20260912T181747Z
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Designation	PILOT / DEMO — SIH 2026 prototype
Officer identity	Not authenticated in current MVP
Settlement	Bhitai Malli
State	Uttarakhand
District	Pauri Garhwal
Coordinates	30.167112, 78.781266 (WGS84 lat, lon)

Important status and disclaimer

- VIKALP is a decision-support platform for identifying and explaining settlement risk — it does not make autonomous government decisions.
- This report presents currently available evidence and the current assessment status; it does not certify a final risk level.
- Pending or unavailable evidence is never interpreted as low risk. Absence of evidence is not evidence of safety.
- Officer review is required before any protection, adaptation, or relocation decision is made.
- This is a pilot/demo assessment for a single settlement (Bhitai Malli, Pauri Garhwal, Uttarakhand) under the SIH 2026 prototype scope.

Settlement Context

Population	383 — Demo planning input — source validation pending
Households	86 — Demo planning input — source validation pending
Elevation	991 m — Demo planning input — source validation pending
Slope (DB/demo value)	18.91° — Demo planning input — source validation pending

No official Census-certified population/household figure is used above; no household-size statistic is calculated or displayed.

Terrain slope discrepancy (disclosed, not resolved): the settlement record's demo slope value is 18.91°, while the separately processed CartoDEM-derived slope at this location is 22.58° (Task 15 raster analysis; not exposed via any current API). VIKALP does not silently choose one value as authoritative. Terrain risk scoring remains pending until this discrepancy and the underlying scoring methodology are resolved through governance review.

GIS / Evidence Sources

A. Administrative boundary context

- geoBoundaries — India ADM2 (district-level administrative boundaries), publisher: geoBoundaries (William & Mary geoLab).
- Filtered to the Uttarakhand subset for this pilot; district-level context only — village-level boundary data is not currently available.

- License: Open Data Commons Open Database License 1.0 (ODbL 1.0), per this dataset's own recorded metadata.

B. Terrain

- ISRO/NRSC Bhuvan CartoDEM v3 R1 (Cartosat-1 Digital Elevation Model) — processed elevation raster and a derived slope raster (Task 14/15).
- Terrain data exists, but terrain risk scoring is currently pending because the scoring governance/data issue (the slope-value discrepancy above) has not been resolved.

C. Landslide evidence

- Geological Survey of India (GSI) / National Landslide Forecasting Centre (NLFC) — “Landslide Inventory (Field Validated)”, Pauri Garhwal pilot extraction (813 records).
- Referred to throughout this report as field-validated landslide inventory evidence — not current landslide probability, not complete landslide risk, and not a real-time landslide forecast.

Risk Assessment Status

Assessment status	pending
Overall score	Not scored
Risk level	Not classified (no overall score)
Data completeness	partial

Overall risk score not computed because the complete five-dimensional assessment is not currently scoreable. This is not a zero score and not a low-risk finding — it is an honest absence of a computable result.

Dimension	Status	Score	Evidence used	Missing inputs
Terrain / Physical Susceptibility	no_scoring_rule	Not scored	Yes	geology, aspect, dem_derived_susceptibility_index
Hazard Exposure	no_evidence_found	Not scored	No	landslide_susceptibility_class, flood_hazard_class, cloudburst_hazard_class, multi_hazard_overlay
Historical Disaster Evidence	no_data	Not scored	No	past_incident_count, past_incident_severity, most_recent_incident_recency
Population / Household Exposure	no_scoring_rule	Not scored	Yes	vulnerable_subgroup_counts
Vulnerability	no_data	Not scored	No	housing_construction_type, distance_to_hospital, distance_to_road, economic_vulnerability_index

Weights shown are prototype policy configuration (see docs/DECISIONS.md), not an official government standard.

Hazard Exposure Evidence

No qualifying GSI landslide inventory record was found within the approved 1 km scoring radius. This means **no qualifying evidence was found within the scoring radius** — it does not mean no hazard exists. VIKALP does not state that Bhitai Malli is safe, and does not state that landslide risk is zero.

Status	no_evidence_found
Qualifying record count	0

Nearest qualifying distance	None within scoring radius
Contextual (1–5 km) record count	21
Nearest contextual distance	2.040 km
Source dataset	GSI/NLFC field-validated landslide inventory, Pauri Garhwal district (Task 20 acquisition, see docs/DATA_PROVENANCE.md)
Source feature count	813

Absence of a GSI inventory record within the scoring radius means no landslide event has been documented and mapped there in this dataset. It does NOT mean this location is safe from landslide hazard (docs/DECISIONS.md Task 21 §11).

Scoring radius, proximity bands, activity modifier, and density bonus are VIKALP prototype policy configuration (docs/DECISIONS.md Task 21/22/23) — NOT an official government risk standard.

Limitations

- Covers only the GSI/NLFC landslide-inventory sub-evidence within Hazard Exposure; flood, cloudburst, and multi-hazard-overlay evidence remain entirely unavailable and are not reflected here.
- The inventory is a record of documented/mapped events, not a complete census of every slope that has failed or could fail — see the inventory-bias disclaimer.
- Distances use a spherical great-circle (haversine) calculation, consistent with the figures already published in Tasks 19-22 — an approximation of true WGS84 ellipsoidal distance, accurate to well under 1% at these ranges.
- This implementation does not distinguish a settlement genuinely outside the acquired inventory's geographic coverage (Pauri Garhwal district) from one inside it with zero nearby records — both currently report "no_evidence_found". A distinct "not_covered" status (Task 21 §12) is not implemented in this MVP.
- Every threshold/band/multiplier above is a VIKALP prototype policy configuration, not a validated or official standard.

Decision Workspace Status

Decision status	pending
Risk assessment status	pending
Officer review required	Yes

Pathway	Status	Recommended
Protect	not_evaluated	No
Adapt	not_evaluated	No
Relocate	not_evaluated	No

VIKALP does not automatically recommend relocation from the current evidence. Incomplete evidence is never treated as a basis for inferring that relocation is warranted.

Evidence required before evaluation

- Completed risk assessment (hazard exposure, historical disaster evidence, vulnerability) is required before any pathway can be evaluated.
- No approved pathway-evaluation rule exists yet for Protect, Adapt, or Relocate — see docs/DECISIONS.md Task 08.

Destination / Capacity Status

Analysis status pending

Ranking status pending

Candidate count 0

Suitability dimensions defined 6

No candidate destination data is currently available for assessment. This is not a statement that no safe destinations exist — no government-curated candidate dataset has been provided to VIKALP yet.

Capacity assessment has not been initiated. No developable land, service, or capacity figure — including zero — is reported for this settlement's relocation destinations.

Data Governance / Provenance

Data / Evidence	Source	Current use	Status / limitation
Administrative boundaries	geoBoundaries India ADM2 (William & Mary geoLab), ODbL 1.0	District-level map context (Uttarakhand)	Village-level boundaries not available
CartoDEM terrain	NRSC/ISRO CartoDEM v3 R1, via Bhuvan (registered-user access)	Source elevation raster for slope derivation	Terrain risk scoring pending governance decision
Derived slope	Derived from CartoDEM via Horn's method (Task 15)	Reference terrain evidence	22.58° derived value vs. 18.91° demo value — discrepancy disclosed, not resolved
GSI landslide inventory	GSI/NLFC Landslide Inventory (Field Validated), Bhusanket portal	Hazard Exposure landslide-proximity evidence	813 records, Pauri Garhwal extraction; documented inventory, not a complete hazard probability model
Population/household demo inputs	Settlement seed record (backend/app/database.py)	Population/Household Exposure raw inputs	Demo planning input — source validation pending; no scoring rule approved yet

Current Limitations

- Pilot geography is one selected region/settlement (Bhitai Malli, Pauri Garhwal, Uttarakhand).
- Full five-dimensional risk scoring is not yet available.
- Terrain scoring is pending.
- Historical disaster scoring is pending.
- Population exposure scoring is pending.
- Vulnerability data is unavailable.
- No government-curated destination candidate dataset is currently integrated.
- Carrying-capacity calculation is pending.
- Relocation recommendation is not produced.
- Current landslide evidence is an inventory, not a complete current hazard probability model.
- Population/household demo inputs require source validation.
- Terrain slope discrepancy remains unresolved.
- Officer review remains necessary.

Officer Review

Officer review status	REQUIRED
Reason	Current evidence is incomplete for a complete multi-dimensional assessment and no autonomous intervention decision is issued.
Officer action	Review available evidence, verify missing inputs, and determine whether further departmental assessment is required.

Officer review / acknowledgement:

Name / designation / signature / date (to be completed by reviewing officer)